



Day 1

Angular Fundamentals



By

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What is a FrontEnd Framework ?





FrontEnd



Template

Style

Logic



Framework

vs

Library

vs

Package

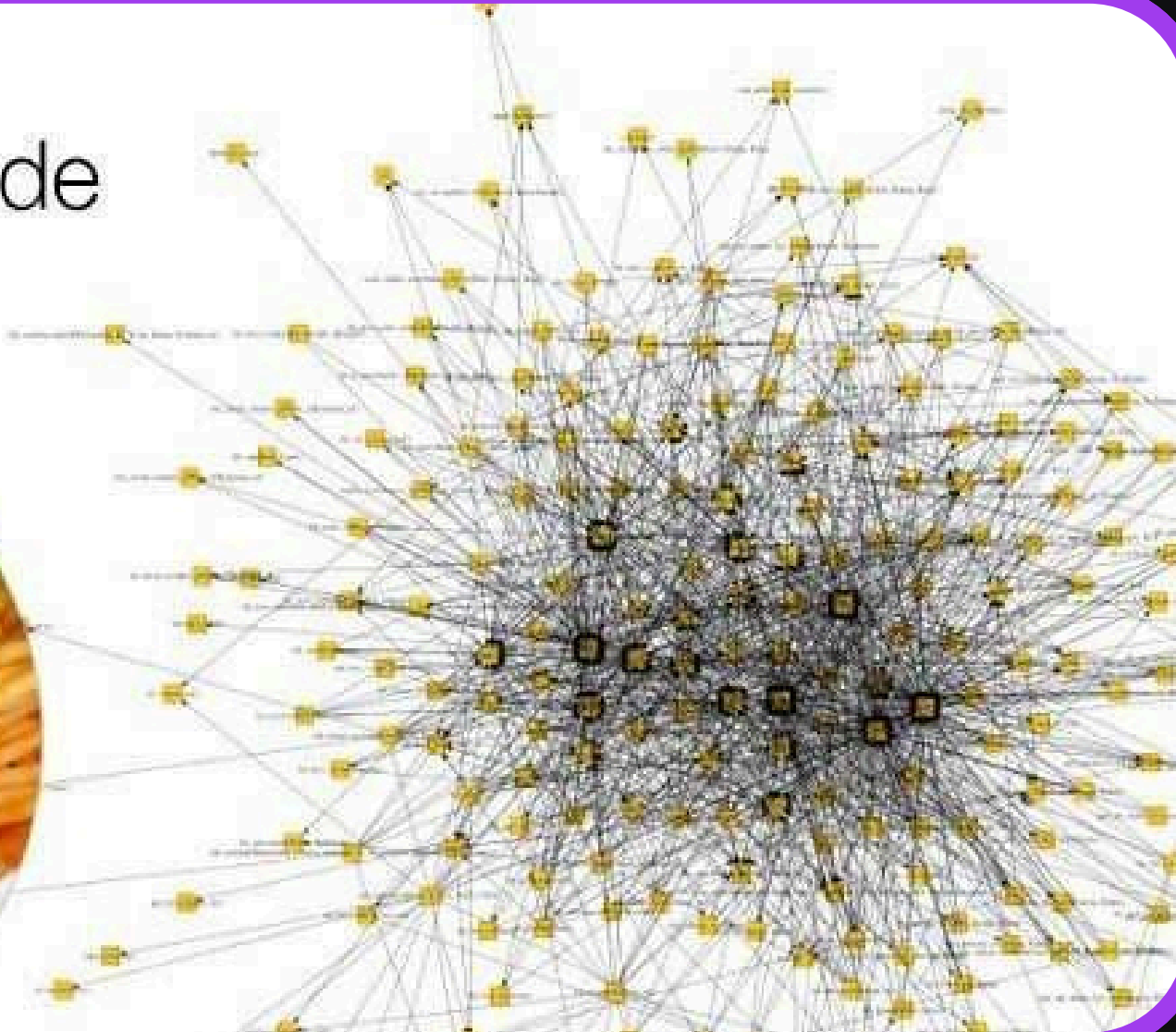
vs

Module



Module

Spaghetti code



Hard Maintenance

Hard To Understand

Duplication

No Reusability

Hard To Test



Module

Functional Programming

All code in a single file

Script

```
Func A() {  
  .....  
}  
Func B() {  
  .....  
}  
Func C() {  
  .....  
}
```

index.js

Code splitted in modules

Script

```
Import A  
Import B  
Import C
```

index.js

Module

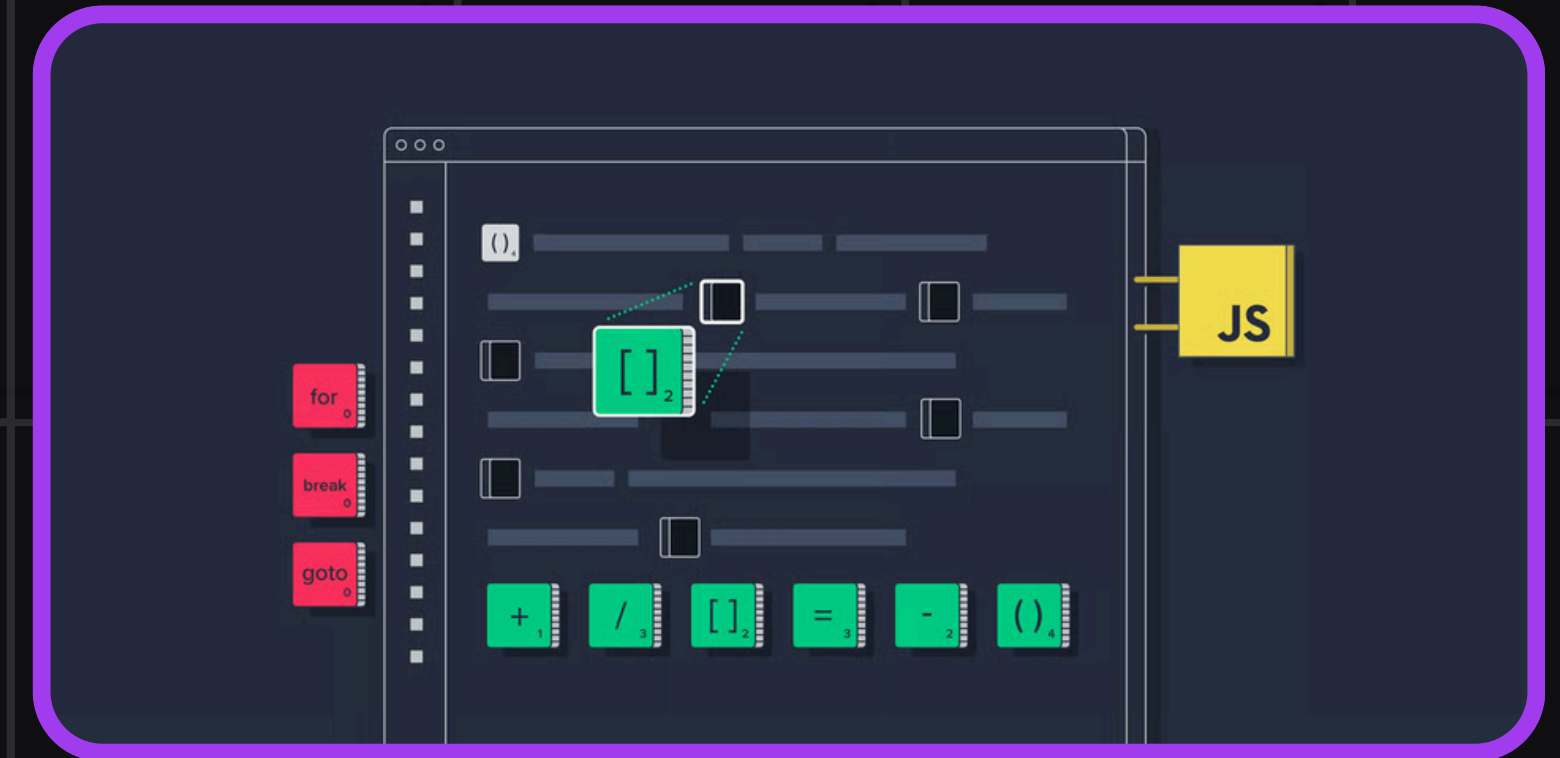
```
Export Func A(){  
  .....  
}
```

Module

```
Export Func B(){  
  Import A  
  .....  
}
```

Module

```
Export Func C(){  
  .....  
}
```





Module



Your fingers



- The smallest piece of software.
- A module is a collection of methods or functions that are ready to be used elsewhere. To reuse the code, modules may be imported into other modules.
- Can be reused in different parts of a project.



Package



Your hands



- A collection of related or not related modules grouped together to perform a task



Library



Building your home



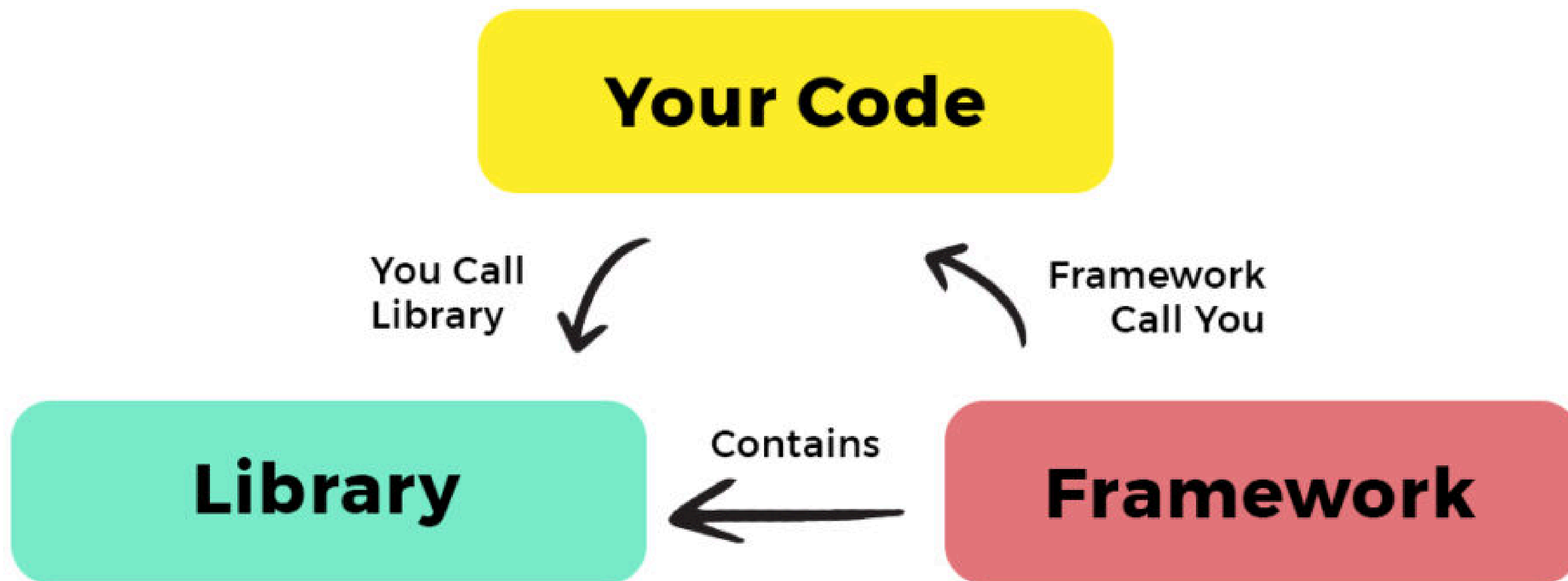
- A library is fundamentally a collection of packages or modules working together to do support specific business. Its objective is to offer a collection of ready-to-use features.
- You control the flow: your code decides when and how to use the library.
- To add functionality to your code, you must include a library.
- no rules and It does not impose any code standards on you as well.
- your code calls the library



Framework

Buying a home

- A complete structure that includes libraries, tools, and rules.
- Provides functionalities and enforce particular design patterns and architectural principles.
- To add functionality to your code, you must include a library.
- The framework controls the flow; your code “plugs into” it.
- the framework calls your code

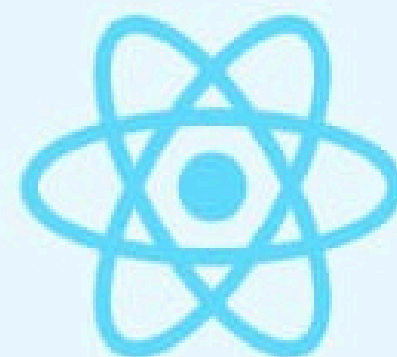




Angular vs React vs Vue



vs



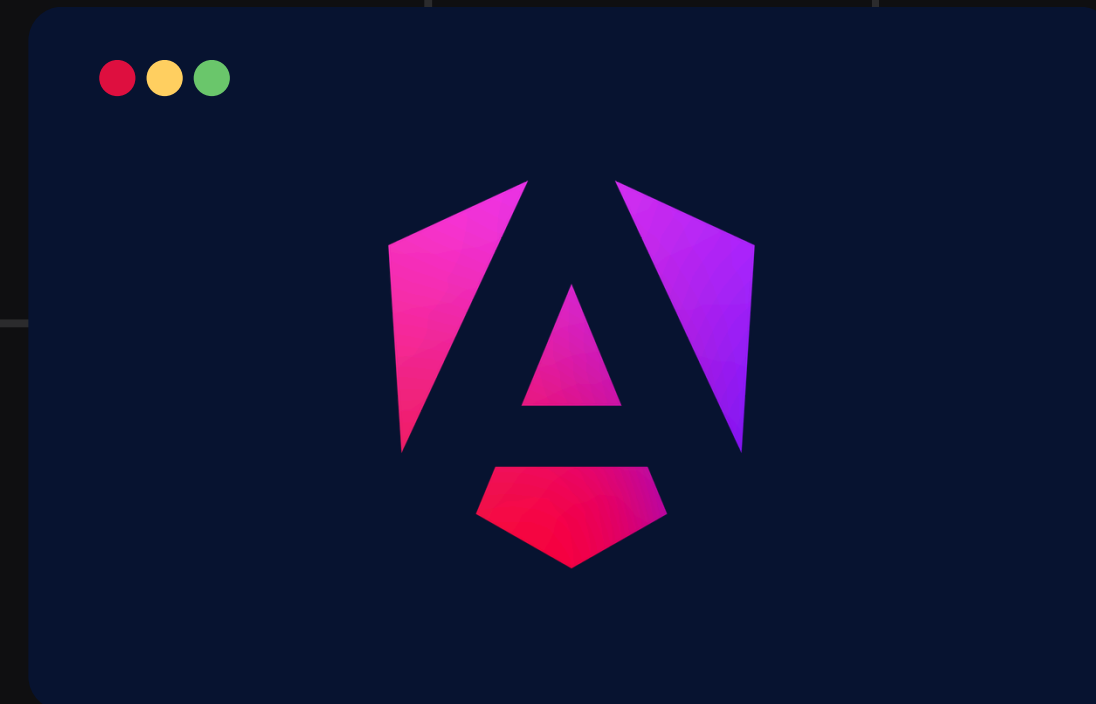
vs



Angular

React

Vue



- frontend framework
- Created in 2010 (AngularJS)
- Developed by Google (Misko Hevery)
- works on real dom



- they were using plain JavaScript + jQuery
- Heavy reliance on DOM manipulation and Imperative coding
- To solve complex front-end logic scattered everywhere
- Lack of structure in large apps

Angular



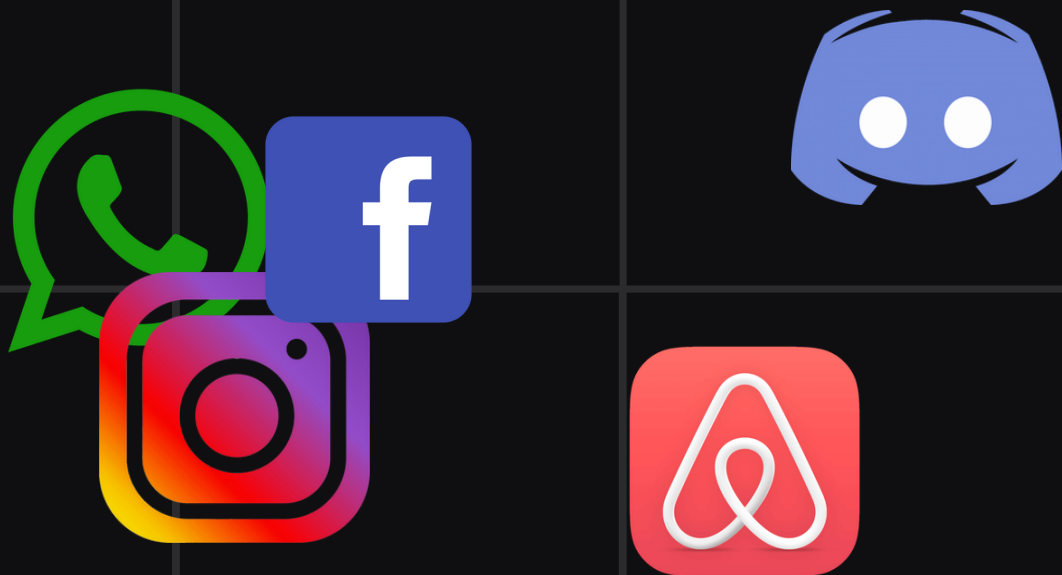


- frontend JavaScript library
- Created in 2013
- Maintained by Meta (Facebook)
- By Jordan Walke
- uses Virtual Dom



- Facebook had complex UI updates using traditional JS
- They struggled with manual DOM manipulation and performance issues

React





- Progressive JavaScript framework
- Created in 2014
- Maintained by Meta (Facebook)
- Invented by Evan You (ex. google engineer)
- Combines ideas from React & Angular



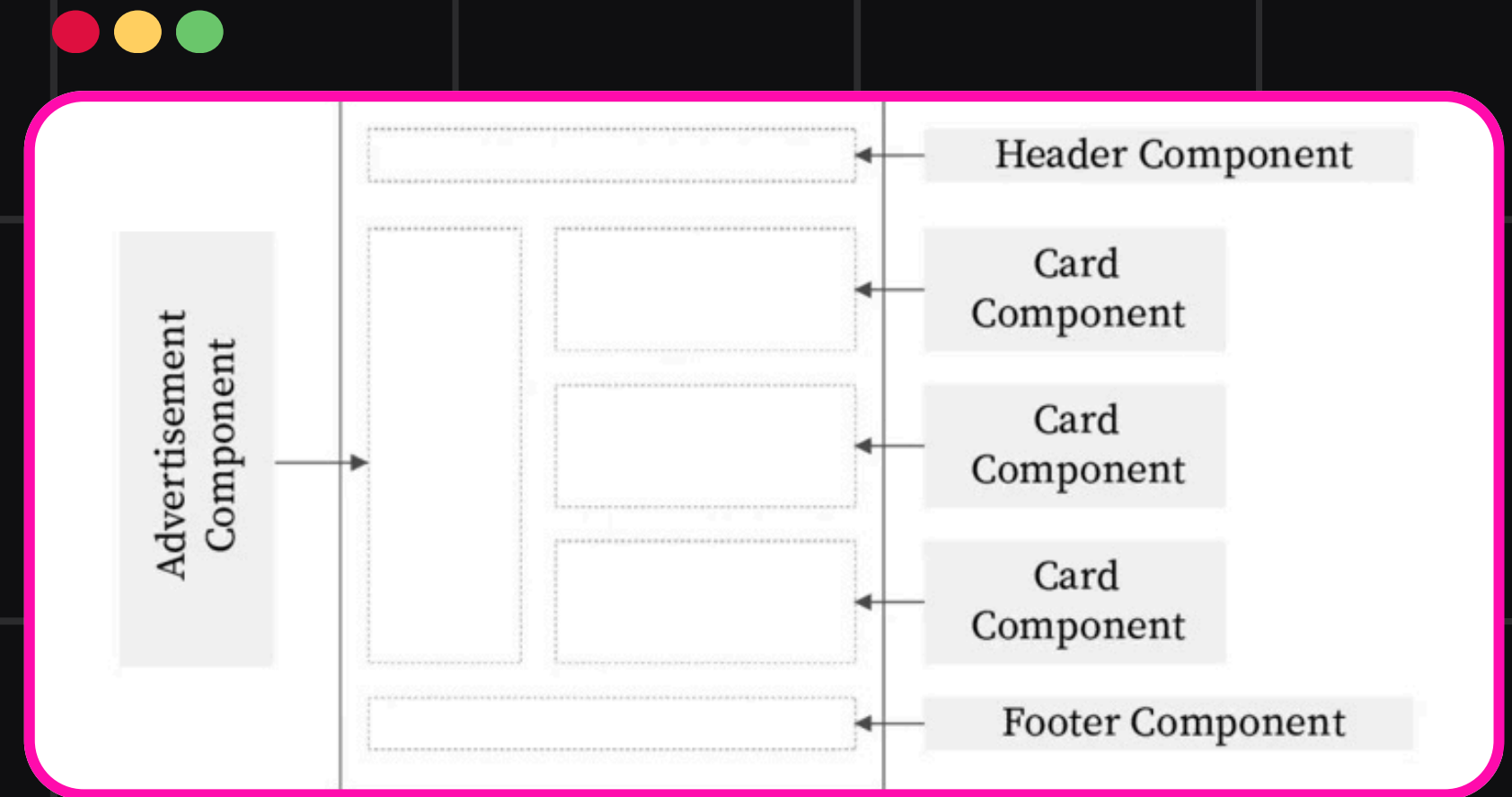
- Found Angular too complex and hard to learn
- create something simpler than Angular and more structured than React

Vue



Component-Based Architecture

break the UI into small independent part



A component is:

A piece of UI (button, navbar, card) has its own:
HTML (template)
CSS (style)
TypeScript (logic)



- Separation
- Reusability
- Maintenance
- Scalability
- Easy Testing





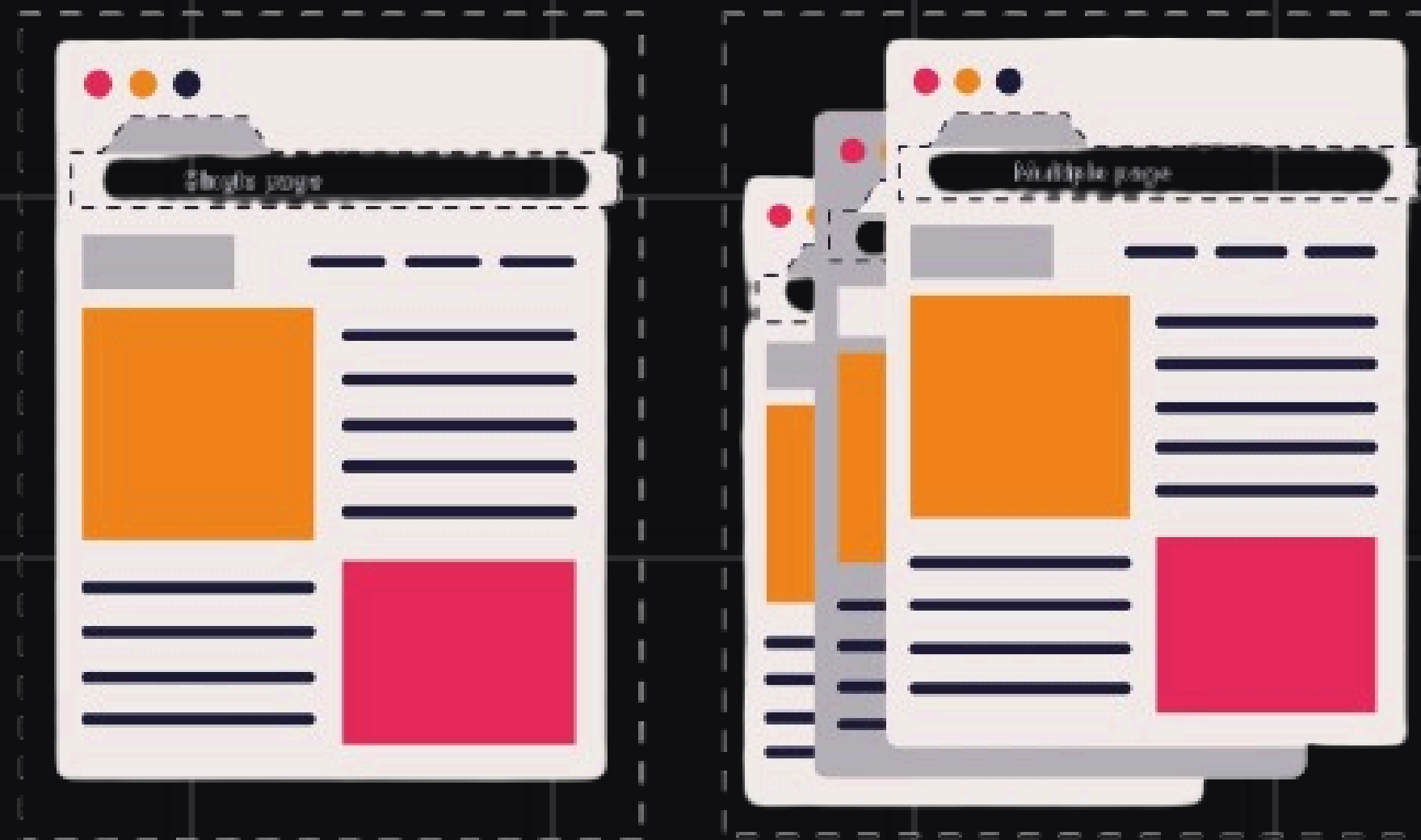
MPA (SSR)

multi page application

VS

SPA (CSR)

single page application



MPA (SSR)



1. Request

Browser asks Server for page



2. Rendering (On Server)

Server fetches data & builds HTML



3. Response

Server sends complete HTML



4. Viewable & Interactive

User sees content immediately



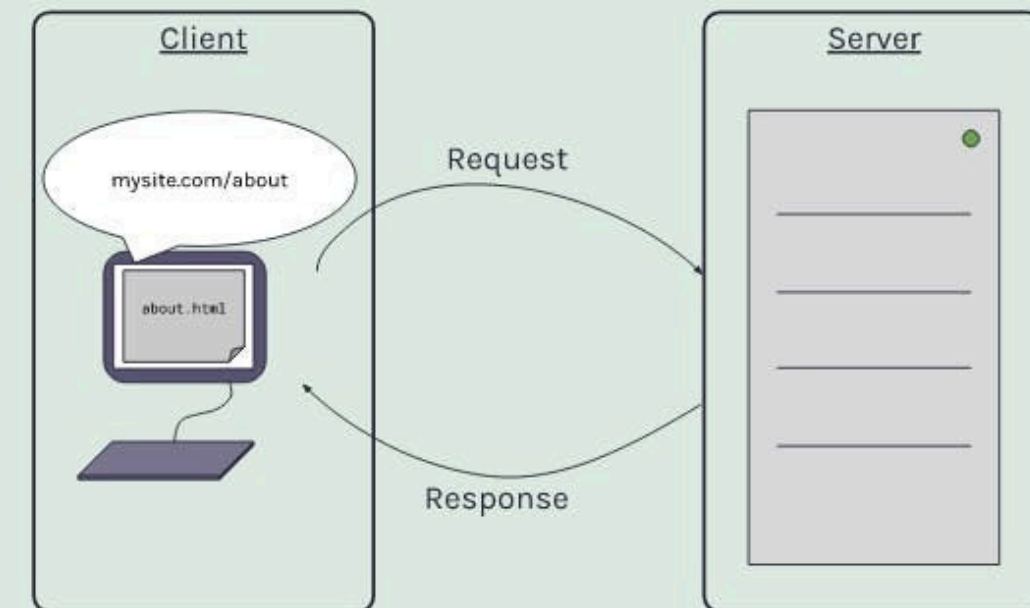
The Server does the heavy lifting. It constructs the full HTML page with all content before sending it to the browser.



The browser receives a fully populated document. The user sees content almost immediately.



Multi-page Apps (MPAs)





MPA (SSR)

The Good

Excellent SEO

Crawlers see full content immediately on the first hit.

Fast Initial Load

First Contentful Paint (FCP) is very quick.

Good for Slow Devices

Less JavaScript work for the user's phone to do.

The Bad

High Server Load

Server renders every page view. Can be expensive.

Slower TTFB

"Time To First Byte" can be slower as server thinks.

Full Page Reloads

Traditional SSR refreshes the whole page on navigation.

Why SEO Loves MPA (SSR)?



- Content is pre-rendered.
- Social Media cards work perfectly (OG Tags).
- No dependency on Googlebot's JS engine capabilities.



```
<html>
  <head>
    <title>Product Name</title>
    <meta name="description" content="Best
product...">
  </head>
  <body>
    <h1>Buy This Now!</h1>
    <p>Detailed description readable by bots...
</p>
  </body>
</html>
```



SPA (CSR)



1. Request

Browser asks Server for page



2. Loading

Server sends blank HTML + JS



3. Rendering

Browser executes JS & builds UI



4. Viewable

User finally sees content



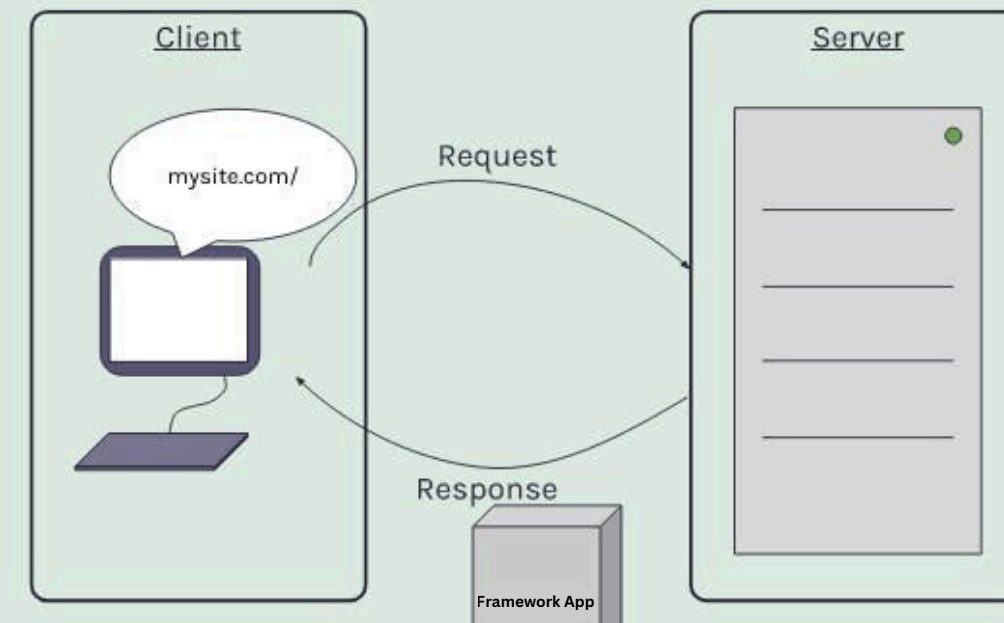
The Browser does the heavy lifting. The server sends a bare-bones HTML page with a JavaScript bundle.



Content is NOT visible immediately. The browser must download and execute JavaScript before the user sees anything meaningful.



Single-page Apps (SPAs) ✨





SPA (CSR)

The Good

Fast Interaction

Once loaded, navigating between pages is instant.

App-like Feel

Smooth transitions and dynamic updates without page refreshes.

Reduced Server Load

Server mainly serves API JSON.

The Bad

Slow Initial Load

User stares at a white screen while JS downloads.

SEO Challenges

Search crawlers might see an empty page.

High Memory Usage

Can be heavy on the user's device battery and RAM.

Why Crawlers Struggle with SPA (CSR)?



Empty Book

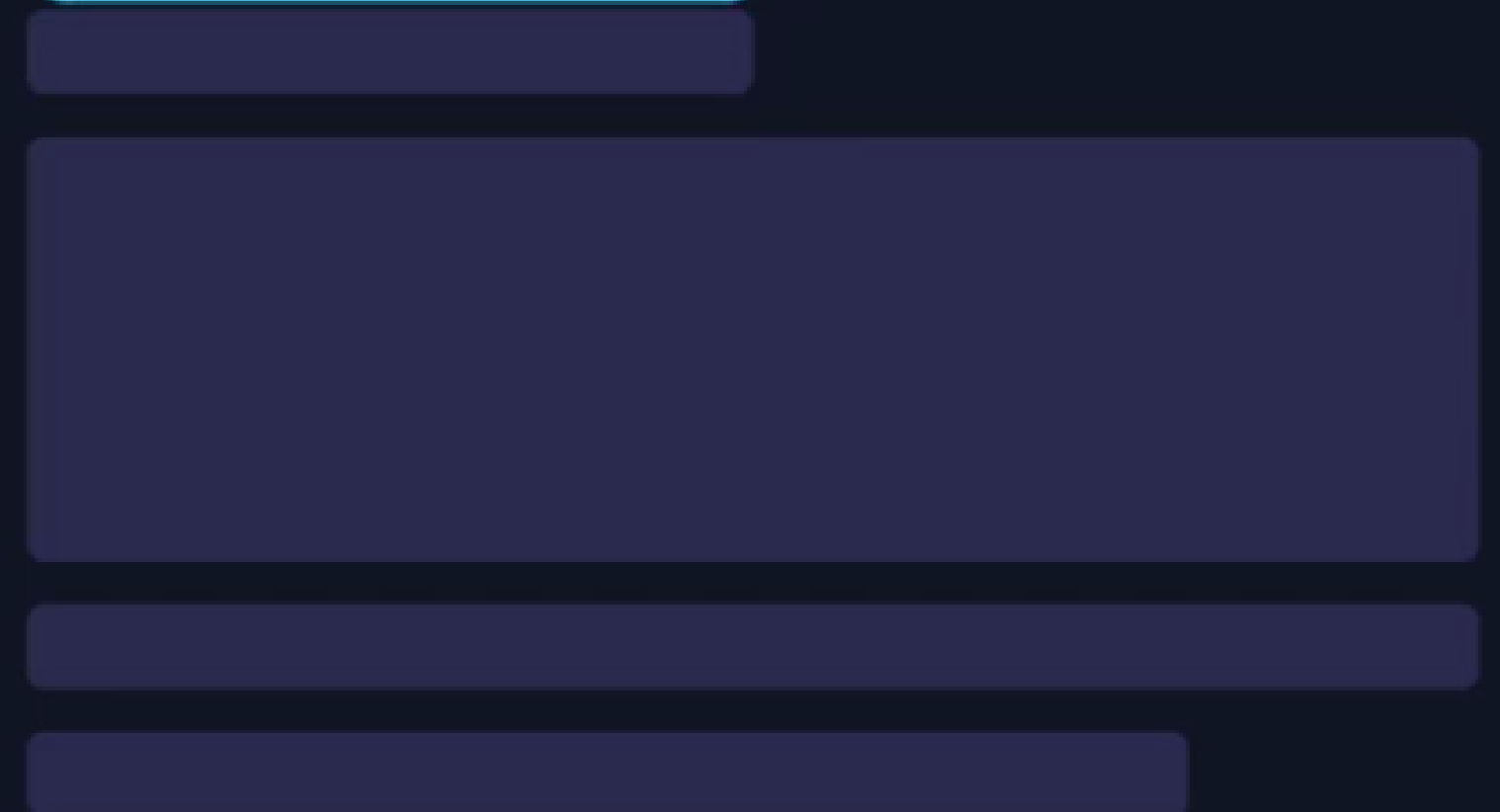
Search engine bots (crawlers) are often fast and impatient. They may not execute JavaScript.

What the Crawler Sees (HTML)

```
<html>
  <body>
    <div id="root"></div>
    <script src="app.js"></script>
  </body>
</html>
```

EMPTY! ❌

What Users See (After JS)



FULL CONTENT ✅



What Is Angular?

Angular is a **frontend JavaScript framework**

It helps with building interactive, modern web user interfaces.

It's also a **collection of tools** & features
CLI, Debugging Tools, IDE Plugins



Why a framework?

Simplifies the process of building complex, interactive web user interfaces

Write **declarative code**

Separation of concerns via **components**

Object-oriented programming concepts & principles

Use **TypeScript**



Declarative vs Imperative

```
</header>

<div id="tabs">
  <menu>
    <app-tab-button
      (select)="onSelectTab(0)"
      [isActive]="activeTab === 0">
      Why Angular?
    </app-tab-button>
    <app-tab-button
      (select)="onSelectTab(1)"
      [isActive]="activeTab === 1">
      Core Features
    </app-tab-button>
    <app-tab-button
      (select)="onSelectTab(2)"
      [isActive]="activeTab === 2">
      Related Resources
    </app-tab-button>
  </menu>
  <div id="tab-content">
    <ul>
      @for (content of activeTabContent; track content) {
        <li>{{ content }}</li>
      }
    </ul>
  </div>
</div>
```

```
for (const item of items) {
  listContent += `<li>${item}</li>`;
}

const list = document.createElement("ul");
tabContent.innerHTML = ""; // clear existing content
list.innerHTML = listContent; // insert new content
tabContent.append(list);
}

function highlightButton(btn) {
  // Clear all existing styling / highlights
  btnWhyAngular.className = "";
  btnCoreFeature.className = "";
  btnResources.className = "";
  btn.className = "active"; // set new style / highlight
}

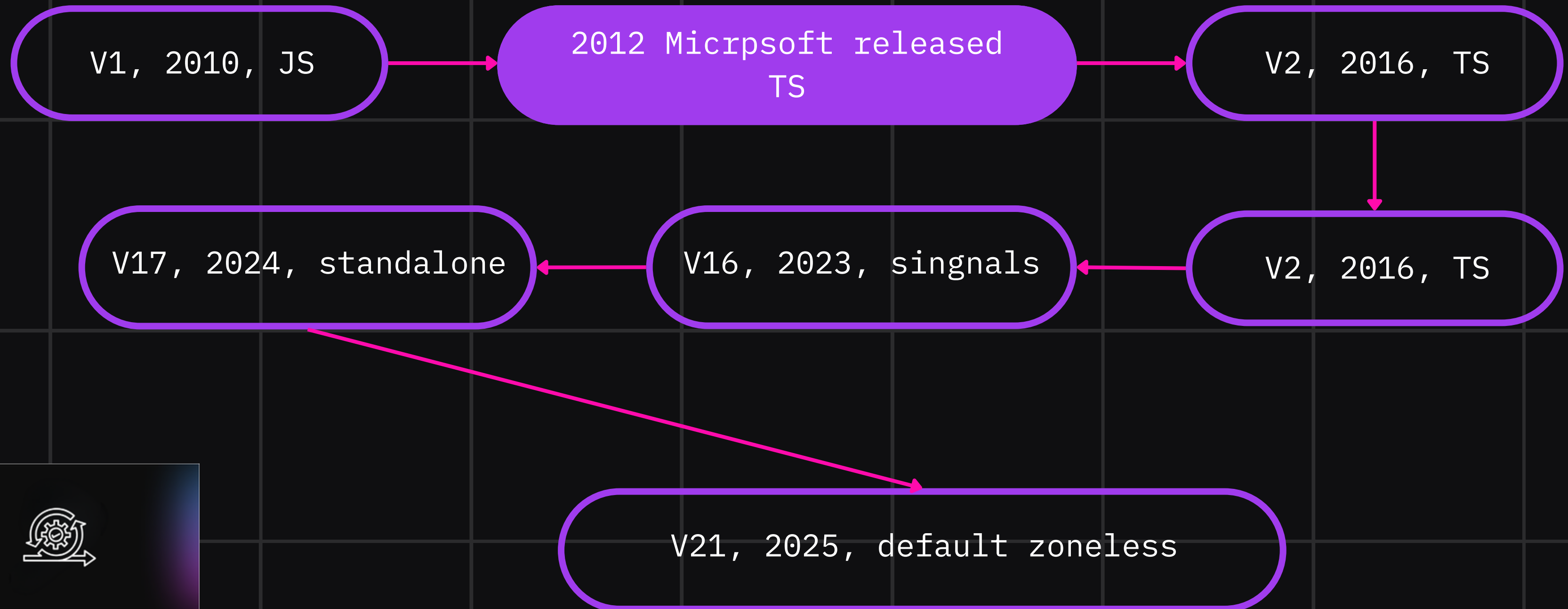
function handleClick(event) {
  const btnId = event.target.id;
  highlightButton(event.target);
  if (btnId === "btn-why-angular") {
    displayContent(content[0]);
  } else if (btnId === "btn-core-features") {
    displayContent(content[1]);
  } else {
    displayContent(content[2]);
  }
}

displayContent(content[0]); // initially show this content
```



Angular Versions History

angular releases a major updates every 6 months



HeroDevs Blog | Angular Version History: Every Release Date, Support Window, and End-of-Life Date from AngularJS to Angular 22

Every Angular version from AngularJS to Angular 22, with release dates, LTS windows, and EOL dates. Learn which versions are still supported, which are silently accumulating unpatched CVEs, and what your options are if you're running...

herodevs.com



Thank You